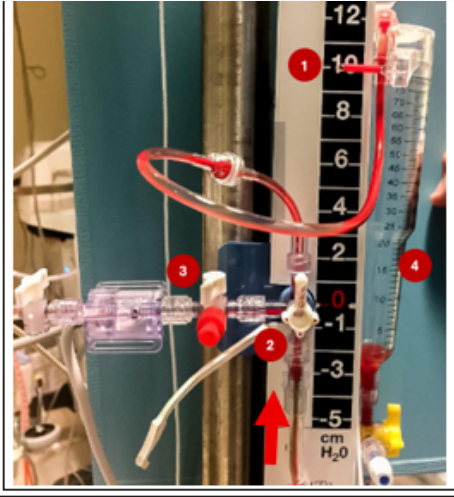


EVD Quick Reference



COMPONENTS

1. **Drainage setting:** Increments of 5 cmH₂O, higher = less drainage (higher resistance).
2. **Drainage stopcock:** 12 o'clock = clamp/closed, 3 o'clock = open to drain.
3. **Transducer and zeroing stopcock:** Controls baseline calibration.
4. **Collection/drip chamber:** Graduated cylinder measuring CSF volume.

*Red arrow indicates the direction of CSF flow.



INDICATIONS

- **CSF diversion** for acute obstructive hydrocephalus (e.g. IVH, posterior fossa stroke/tumor).
- **Intracranial Pressure (ICP) monitoring** in patients with severe brain injury (GCS $\leq$ 8).

SIGNS OF HYDROCEPHALUS

Clinical Signs:

- Decline in Level of Consciousness (LOC) or progressive somnolence.
- **Parinaud's Syndrome:** Upward gaze palsy (setting sun sign), retraction nystagmus on convergence, and pupillary light-near dissociation.

Radiographic Signs (NCCT Head):

- Progressive enlargement of the cerebral ventricles.
- Temporal horn dilation (sensitive early sign of obstruction).
- High-risk factors: Intraventricular Hemorrhage (IVH) in 3rd or 4th ventricles, compression of 4th ventricle, or high volume blood (mGS $\geq$ 6).

BASICS

- **Leveling:** Always align the zero level of the EVD scale/transducer to the external auditory meatus (EAM) / tragus.
- **Repositioning:** Clamp the EVD prior to any changes in Head of Bed (HOB) angle, patient transfers, or physical therapy.
- **Waveform Validity:** ICP value and waveform morphology are valid only when the EVD is clamped.
- **CSF Drainage:** CSF will drain only when the patient's actual ICP exceeds the set drainage height setting.

COMPLICATIONS

- **Overdrainage (>20 mL/hr):** Risk of subdural hematomas (bridging vein tearing), ventricular collapse (slit ventricles), or upward cerebellar herniation.
- **Underdrainage:** Risk of worsening hydrocephalus, brain compression, or elevated ICP. Troubleshoot for system kinks, blood clots, air locks, or malpositioned stopcocks.